

math111a hw1

jc — October 02, 2025

Problem 1

Let m and n be **coprime** positive integers. Prove that the function

$$\mathbb{Z}/mn\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$$

$$a \bmod mn \mapsto (a \bmod m, a \bmod n)$$

is well defined and bijective. (This result is commonly known as the Chinese Remainder Theorem).

Solution:

Well defined: suppose that $a \equiv b \bmod mn$ then according to the function definition:

$$a \bmod mn \mapsto (a \bmod m, a \bmod n)$$

$$b \bmod mn \mapsto (b \bmod m, b \bmod n)$$

Since $a \equiv b \bmod mn$ then we have $mn \mid a - b$ so that $m \mid a - b$ and $n \mid a - b$ so that $a \equiv b \bmod m$ and $a \equiv b \bmod n$ and the function is well defined (the output does not depend on the choice of representative for $a \bmod mn$)

Injective: Suppose that $(a \bmod m, a \bmod n) = (b \bmod m, b \bmod n)$ then $a \equiv b \bmod m$ and $a \equiv b \bmod n \Rightarrow m \mid a - b$ and $n \mid a - b$

Since m, n are coprime then their LCM mn must divide $a - b$ i.e. $a \equiv b \bmod mn$ (see problem 2 in the reference)

Surjective: Note that $\mathbb{Z}/mn\mathbb{Z}$ has mn elements. Also note that $|\mathbb{Z}/m\mathbb{Z}| = m$ and $|\mathbb{Z}/n\mathbb{Z}| = n$ so that $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ also has mn elements. Since the domain and codomain have the same cardinality we can conclude that the function is surjective.

■

References:

https://web.math.ucsb.edu/~marcosreyes/teaching/Fall2021-8/Math_8_Quiz_5_Solutions.pdf

Problem 2

Which of the following sets are closed under the given operation?

- a) $\{0, 4, 8, 12\}$ under addition modulo 16
- b) $\{0, 4, 8, 12\}$ under addition modulo 15
- c) $\{1, 4, 7, 13\}$ under multiplication modulo 15
- d) $\{1, 4, 5, 7\}$ under multiplication modulo 9

Solution:

- a) This set is closed since we are adding two integers of the form $4n$ where $n \in \{0, 1, 2, 3\}$ and then taking the remainder modulo 16, which is another multiple of 4
- b) Not closed since $4 + 12 \bmod 15 = 1 \notin \{0, 4, 8, 12\}$
- c) Closed since all possible multiplications between elements result in another element in the set
- d) Not closed since $4 * 5 \bmod 9 = 2 \notin \{1, 4, 5, 7\}$

Problem 3

- a) Let n and a be positive integers and $\gcd(n, a) = d$. Show that the equation $ax \bmod n = 1$ has a solution iff $d = 1$. (Essentially what this exercise says is that an integer a has a multiplicative inverse modulo n iff a and n are coprime.)
- b) Define $(\mathbb{Z}/n\mathbb{Z})^\times$ to be the set of all positive integers less than n and relatively prime to n . List out all the elements of $(\mathbb{Z}/20\mathbb{Z})^\times$
- c) The set $(\mathbb{Z}/n\mathbb{Z})^\times$ is a group under multiplication modulo n (no need to prove this). Find the inverse of the elements 2, 7, and 8 in $(\mathbb{Z}/9\mathbb{Z})^\times$

Solution:

a)

$\Rightarrow$ Suppose that $ax \equiv 1 \pmod{n}$ then $n \mid ax - 1 \Rightarrow ax - 1 = kn \Rightarrow ax - kn = 1$ for some $k \in \mathbb{Z}$

Since a, n are positive, we can apply Bezout's identity directly: by assumption there exist integers x and $-k$ such that the above holds and conclude that $\gcd(n, a) = 1$

$\Leftarrow$ Suppose that $\gcd(n, a) = 1$ then by Euler's theorem we have that $a^{\phi(n)} \equiv 1 \pmod{n}$ and so a has the multiplicative inverse $a^{\phi(n)-1}$ (denoted x in the problem statement)

b) $(\mathbb{Z}/20\mathbb{Z})^\times = \{1, 3, 7, 9, 11, 13, 17, 19\}$

c) The inverses are 5, 4, and 8 respectively

Since the modulus is small, we can brute force find these but in general the extended euclidian algorithm can be used

Sources: https://en.wikipedia.org/wiki/Euler%27s_theorem

https://proofwiki.org/wiki/B%C3%A9zout%27s_Identity

Problem 4

Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$

- prove that G is a group under multiplication. This group is often called the group of roots of unity in $\mathbb{C}$.
- prove that G is not a group under addition

Solution:

The group of roots of unity G is also equal to

$$G = \left\{ \cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right) : k \in \{0, 1, \dots, n-1\} \right\}$$

For $k = 0$ we have $\cos(0) + i \sin(0) = 1 = \left(\cos\left(\frac{2\pi}{n}\right) + i \sin\left(\frac{2\pi}{n}\right)\right)^0$

For $k = 1$ we have $\cos\left(\frac{2\pi}{n}\right) + i \sin\left(\frac{2\pi}{n}\right)$

For $k = 2$ we have $\cos\left(\frac{2 \cdot 2\pi}{n}\right) + i \sin\left(\frac{2 \cdot 2\pi}{n}\right) = \left(\cos\left(\frac{2\pi}{n}\right) + i \sin\left(\frac{2\pi}{n}\right)\right)^2$ by De Moivre's formula

Then for $k = n-1$ we obtain $\left(\cos\left(\frac{2\pi}{n}\right) + i \sin\left(\frac{2\pi}{n}\right)\right)^{n-1}$

If we let $r = \cos\left(\frac{2\pi}{n}\right) + i \sin\left(\frac{2\pi}{n}\right)$ then we can write $G = \{r^0, r^1, r^2, \dots, r^{n-1}\}$

a)

Closure: let $k_1, k_2 \in \{0, 1, \dots, n-1\}$ and consider the value $r^{k_1} r^{k_2}$

Since $r \in \mathbb{C}$ we have that $r^{k_1} r^{k_2} = r^{k_1+k_2}$ and we consider three cases:

$k_1 + k_2 < n$: then clearly $r^{k_1} r^{k_2} \in G$

$k_1 + k_2 = n$: then $r^{k_1+k_2} = r^n = \left(\cos\left(\frac{2\pi}{n}\right) + i \sin\left(\frac{2\pi}{n}\right)\right)^n = \cos(2\pi) + i \sin(2\pi) = 1 + i(0) = 1 \in G$

$k_1 + k_2 > n$: by the division theorem $k_1 + k_2 = nq + t$ where $0 \leq t < n$ and so we obtain

$$r^{k_1+k_2} = r^{nq+t} = (r^n)^q r^t = (1)^q r^t = r^t \in G$$

Associativity: let $r^{k_1}, r^{k_2}, r^{k_3} \in G$ where $k_1, k_2, k_3 \in \{0, 1, \dots, n-1\}$ then $(r^{k_1} r^{k_2}) r^{k_3} = r^{k_1} (r^{k_2} r^{k_3})$ by associativity of complex numbers

Identity: the identity is the element $r^0 = 1 \in G$. Then by identity property of complex numbers then for any $a \in G$ we will have that $a \cdot 1 = a = 1 \cdot a$

Inverse: the inverse of an element r^k where $k \in \{0, 1, \dots, n-1\}$ is r^{n-k} since $r^k r^{n-k} = r^{k+n-k} = r^n = 1 \in G$

Therefore all group properties hold under complex multiplication. ■

b) Consider $r^0 + r^0 = 1 + 1 = 2$ and that no $n \in \mathbb{Z}^+$ can satisfy $2^n = 1$. Therefore G is not closed under addition and also not a group under addition ■

Sources:

https://en.wikipedia.org/wiki/Root_of_unity#Group_of_all_roots_of_unity

https://en.wikipedia.org/wiki/De_Moivre's_formula

Problem 5

Let $G = \{a + b\sqrt{2} \in \mathbb{R} \mid a, b \in \mathbb{Q}\}$

- Prove that G is a group under addition
- Prove that the nonzero elements of G form a group under multiplication

Solution:

Let $x = a + b\sqrt{2}$, $y = c + d\sqrt{2}$, and $z = e + f\sqrt{2}$ where $a, b, c, d, e, f \in \mathbb{Q}$ and $x, y, z \in G$ be arbitrary elements in G

- Closure: let x, y be any two elements in G then we have that

$$x + y = a + b\sqrt{2} + c + d\sqrt{2} = (a + b) + (c + d)\sqrt{2} \in G$$

Associativity: $(x + y) + z = (a + b\sqrt{2} + c + d\sqrt{2}) + e + f\sqrt{2} = a + b\sqrt{2} + (c + d\sqrt{2} + e + f\sqrt{2}) = x + (y + z)$ by associativity of addition in the real numbers

Identity: The identity element is $e = 0 + 0\sqrt{2} = 0$. It follows that for any element $x \in G$ that $x + e = x = e + x$

Inverse: the inverse of any element $x = a + b\sqrt{2} \in G$ denoted x^{-1} is $-a - b\sqrt{2}$ and it is clear that for any $x \in G$ that we have $x + x^{-1} = a + b\sqrt{2} - a - b\sqrt{2} = 0 = e = 0 = -a - b\sqrt{2} + a + b\sqrt{2} = x^{-1} + x$ following the usual rules for addition of real numbers

Since all the group properties hold then G is a group ■

- Closure: let x, y be any two elements in G then

$$xy = (a + b\sqrt{2})(c + d\sqrt{2}) = ac + \sqrt{2}ad + \sqrt{2}bc + 2bd = (ac + 2bd) + ((ad + bc)\sqrt{2}) \in G$$

Associativity:

$$(xy)z = ((a + b\sqrt{2})(c + d\sqrt{2}))(e + f\sqrt{2}) = ac + \sqrt{2}ad + \sqrt{2}bc + 2bd(e + f\sqrt{2}) = eac + \sqrt{2}acf + \sqrt{2}ead + 2adf + \sqrt{2}ebc + 2bcf + 2ebd + 2\sqrt{2}bdf$$

$$x(yz) = (a + b\sqrt{2})((c + d\sqrt{2})(e + f\sqrt{2})) = (a + b\sqrt{2})ec + \sqrt{2}cf + \sqrt{2}ed + 2df = eac + \sqrt{2}acf + \sqrt{2}ead + 2adf + \sqrt{2}ebc + 2bcf + 2ebd + 2\sqrt{2}bdf$$

and so both sides are equal

Identity: the identity is $e = 1 + 0\sqrt{2} = 1$. Then since any $x \in G$ is a real number it follows that

$$xe = x(1) = x = 1(x) = ex$$

Inverse: 0 cannot have an inverse since $r(0) = 0 \neq e$ for any $r \in \mathbb{R}$.

For non-zero elements of G the inverse x^{-1} for an arbitrary $x \in G$ is $\frac{1}{a+b\sqrt{2}}$. Then $xx^{-1} = 1 = x^{-1}x$

We can check that the claimed x^{-1} is in fact an element in G :

$$\frac{1}{a + b\sqrt{2}} = \frac{1}{a + b\sqrt{2}} \cdot \frac{a - b\sqrt{2}}{a - b\sqrt{2}} = \frac{a - b\sqrt{2}}{a^2 - 2b^2} = \left(\frac{a}{a^2 - 2b^2}\right) + \left(\frac{-b}{a^2 - 2b^2}\right)\sqrt{2} \in G$$

Since all group properties hold if we exclude 0, G is a group under multiplication ■

Problem 6

Let x and y be elements of a group G .

Prove that $xy = yx$ if and only if $y^{-1}xy = x$ if and only if $x^{-1}y^{-1}xy = 1$

Solution:

We can prove this statement in three parts (let 1 be the identity element of G)

a) $xy = yx \Rightarrow y^{-1}xy = x$

Left multiply both sides by y^{-1} : $y^{-1}(xy) = y^{-1}(yx) \Rightarrow y^{-1}xy = (y^{-1}y)x = 1x = x$ by associativity and existence of inverse in groups

b) $y^{-1}xy = x \Rightarrow x^{-1}y^{-1}xy = 1$

Left multiply both sides by x^{-1} : $x^{-1}(y^{-1}xy) = x^{-1}x \Rightarrow x^{-1}y^{-1}xy = 1$

c) $x^{-1}y^{-1}xy = 1 \Rightarrow xy = yx$

$x(x^{-1}y^{-1}xy) = x(1)$	left multiply by x
$y^{-1}xy = x$	associativity, inverse, identity properties
$y(y^{-1}xy) = y(x)$	left multiply by y
$xy = yx$	associativity, inverse, identity properties

■